Microsoft Elevate

CAPSTONE PROJECT

# Sleep Health and Lifestyle Analytics: A Data-Driven Study of Stress, Activity, and Sleep Disorders


PRESENTED BY

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# OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Result
- Conclusion
- Future Scope
- References

# PROBLEM STATEMENT:

Sleep is essential for maintaining physical and mental health, but modern lifestyle factors such as high stress, irregular work schedules, and low physical activity are negatively affecting sleep quality. Many people experience sleep disorders like insomnia and sleep apnea without understanding how factors such as occupation, stress levels, BMI, and daily habits influence their sleep patterns.

Analyzing raw health data can be difficult without proper visualization tools. Therefore, this project develops an interactive Power BI dashboard to analyze sleep duration, sleep quality, stress levels, BMI, physical activity, and sleep disorders. The dashboard helps identify patterns and insights that explain how lifestyle factors impact sleep health.

# PROPOSED SOLUTION:

The proposed solution uses data analytics and visualization techniques to analyze the relationship between lifestyle factors and sleep health.
An interactive Power BI dashboard is developed to explore how variables such as sleep duration, stress level, BMI, occupation, physical activity, and daily steps influence sleep quality and sleep disorders.

The system provides clear visual insights that help identify patterns, correlations, and potential risk factors affecting sleep health.

**Key Components of the Solution**

- Data Collection
- Data Cleaning and Preparation
- Data Analysis and Visualization
- Interactive Dashboard Development
- Insight Generation

# PROPOSED SOLUTION:

**Data Collection**

The Sleep Health and Lifestyle dataset is collected containing information such as:

- Age and Gender
- Occupation
- Sleep Duration
- Quality of Sleep
- Stress Level
- BMI Category
- Physical Activity Level
- Daily Steps
- Sleep Disorders

**Data Preparation**

The dataset is imported into Power BI.

Data cleaning is performed to:

- Remove missing or inconsistent values
- Standardize categorical data

- Data transformation and formatting are applied to prepare the dataset for analysis.

# PROPOSED SOLUTION:

**Data Analysis**

Different lifestyle factors are analyzed to understand their impact on sleep health.

Key relationships analyzed include:

- Sleep Duration vs Occupation
- Stress Level vs Sleep Disorder
- BMI Category vs Sleep Disorder
- Physical Activity vs Sleep Quality
- Gender vs Sleep Health

**Visualization Development**

Various Power BI visuals are created to represent insights:

- Bar Charts
- Pie Charts
- KPI Cards
- Line Charts
- Stacked Column Charts
- Filters & Slicers

These visualizations help users easily explore and interpret sleep health patterns.

# PROPOSED SOLUTION:

**Dashboard Development**

An interactive Power BI dashboard is created that includes:

- Sleep health overview metrics
- Gender-based sleep analysis
- Occupation-based sleep patterns
- Sleep disorder analysis
- BMI and lifestyle factor analysis
- Interactive filters and slicers allow users to analyze data dynamically by gender, occupation, and sleep disorder type.

**Insights and Decision Support**

The dashboard helps:

- Identify occupations with higher stress levels
- Understand how BMI and physical activity influence sleep quality
- Detect patterns related to insomnia and sleep apnea
- Provide data-driven insights for improving sleep health awareness.

# SYSTEM APPROACH:

The system approach describes the methodology used to analyze sleep health data using Power BI. The approach focuses on collecting the dataset, preparing the data, and creating interactive visualizations to explore relationships between lifestyle factors and sleep patterns.

**System Requirements**

**Hardware Requirements**

- Processor: Intel i3 / i5 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: 500 MB or more free space

**Software Requirements**

- Microsoft Power BI Desktop
- Microsoft Excel / CSV dataset
- Windows Operating System

# SYSTEM APPROACH:

**Dataset Used: Sleep Health and Lifestyle Dataset**

**Dataset Description:**

This dataset contains information about individuals' sleep habits and lifestyle factors that influence sleep health. It helps analyze the relationship between sleep duration, stress levels, physical activity, BMI, occupation, and sleep disorders.

**Key Attributes in the Dataset**

| | |
|---|---|
| Age | Stress Level |
| Gender | BMI Category |
| Occupation | Physical Activity Level |
| Sleep Duration | Daily Steps |
| Quality of Sleep | Heart Rate |
| Sleep Disorder | Blood Pressure |

**Purpose of the Dataset**

The dataset is used to analyze how different lifestyle factors affect sleep patterns and to identify trends related to sleep quality and sleep disorders.

# SYSTEM APPROACH:

**Power BI Features Used:**

**Slicers:** These allow users to dynamically filter the dashboard data.

- Gender slicer
- Occupation slicer
- Sleep disorder filter

**KPI Cards:** Used to display key metrics such as:

- Average Sleep Duration
- Average Stress Level
- Average Physical Activity Level

**Bar Charts**

- Stress level by occupation
- Sleep duration by occupation

# SYSTEM APPROACH:

**Power BI Features Used:**

**Pie Charts:** Used to represent:

- Gender comparison in sleep duration and quality
- BMI category distribution

**Line / Combined Charts:** Used to compare:

- Daily steps and physical activity level across occupations

**Stacked Column Charts:** Used to analyze:

- Sleep disorders by occupation
- BMI category distribution across occupations

# RESULT:

The developed Power BI dashboard successfully visualizes the relationship between sleep health and lifestyle factors such as stress level, physical activity, BMI, occupation, and sleep disorders. The interactive dashboard enables users to explore patterns and trends in sleep duration and sleep quality.

**Key Results**

- Identification of occupations with higher stress levels and lower sleep quality.
- Analysis of the relationship between physical activity and sleep duration.
- Visualization of BMI category distribution and its impact on sleep disorders.
- Comparison of sleep health patterns across gender and occupations.

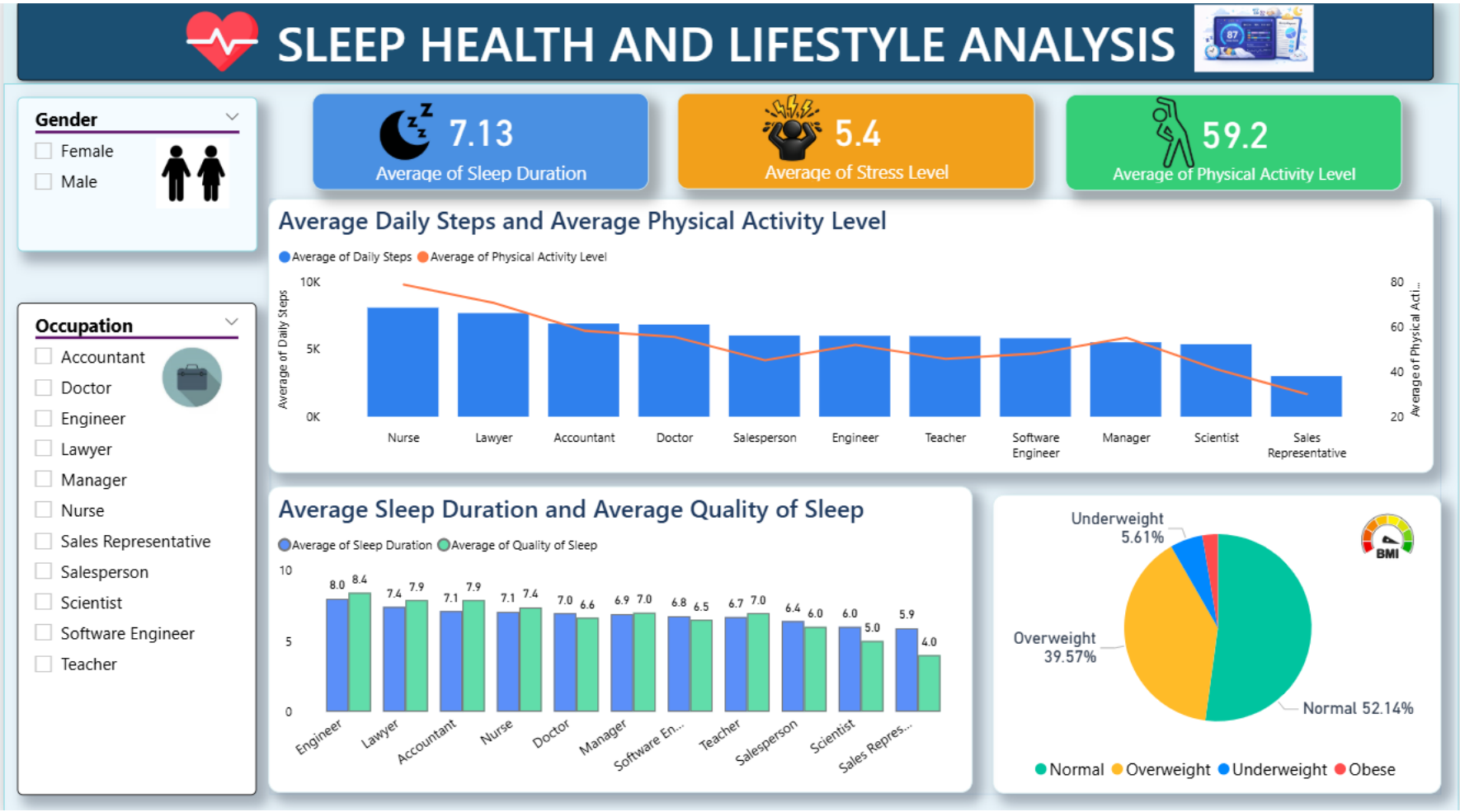
**Project Resources**

- **GitHub Repository:** [Project Repository Link]
- **Google Drive:** [Project Files Link]

# RESULT:

**Sleep Health Overview:**

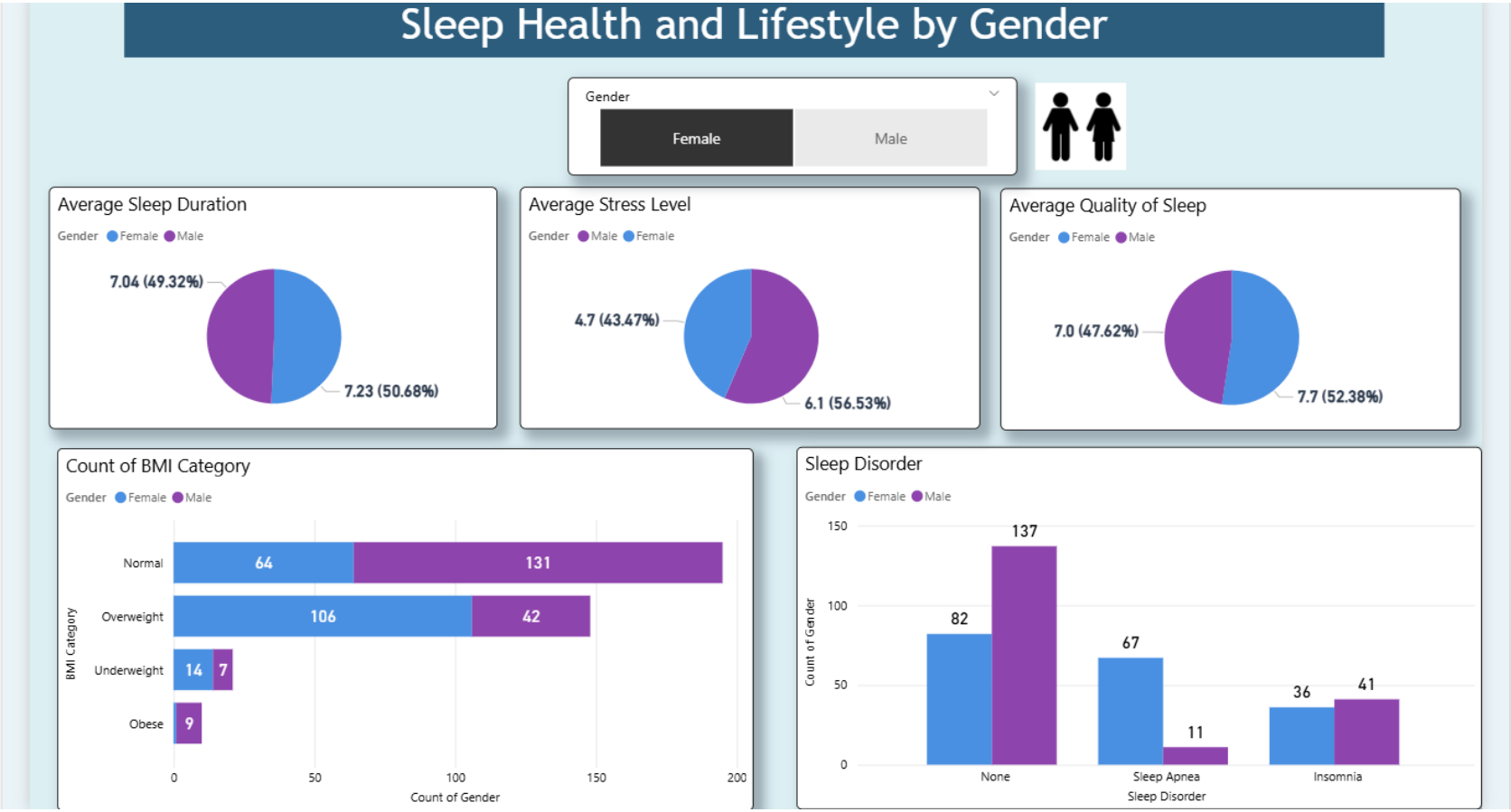
Overview dashboard showing sleep health, stress, activity, and BMI insights. [Link]



# RESULT:

**Sleep Health Analysis By Gender:**

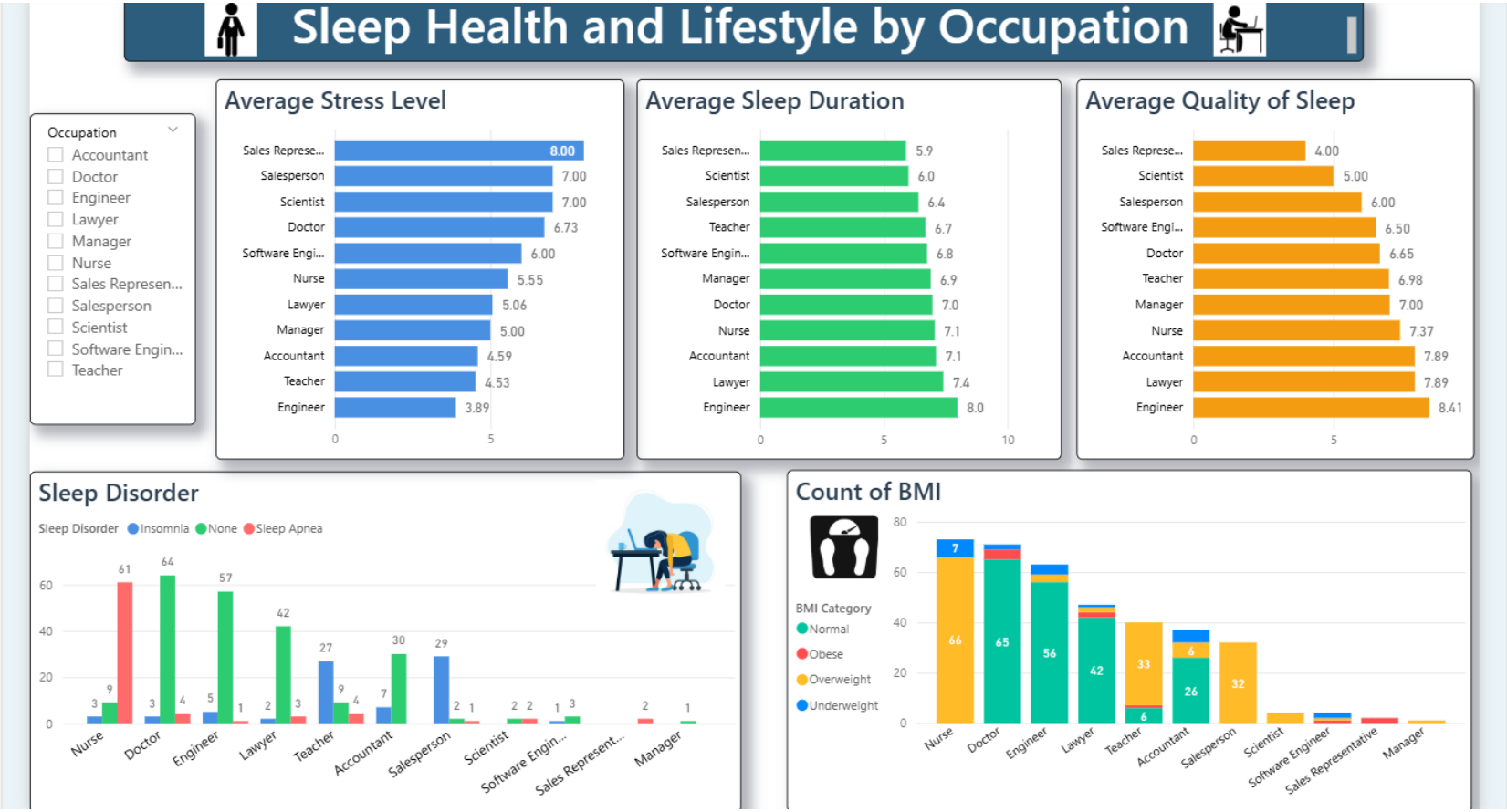
Gender-based comparison of sleep duration, stress levels, sleep quality, BMI, and sleep disorders. [Link]



# RESULT:

**Sleep Health and Lifestyle by Occupation:**

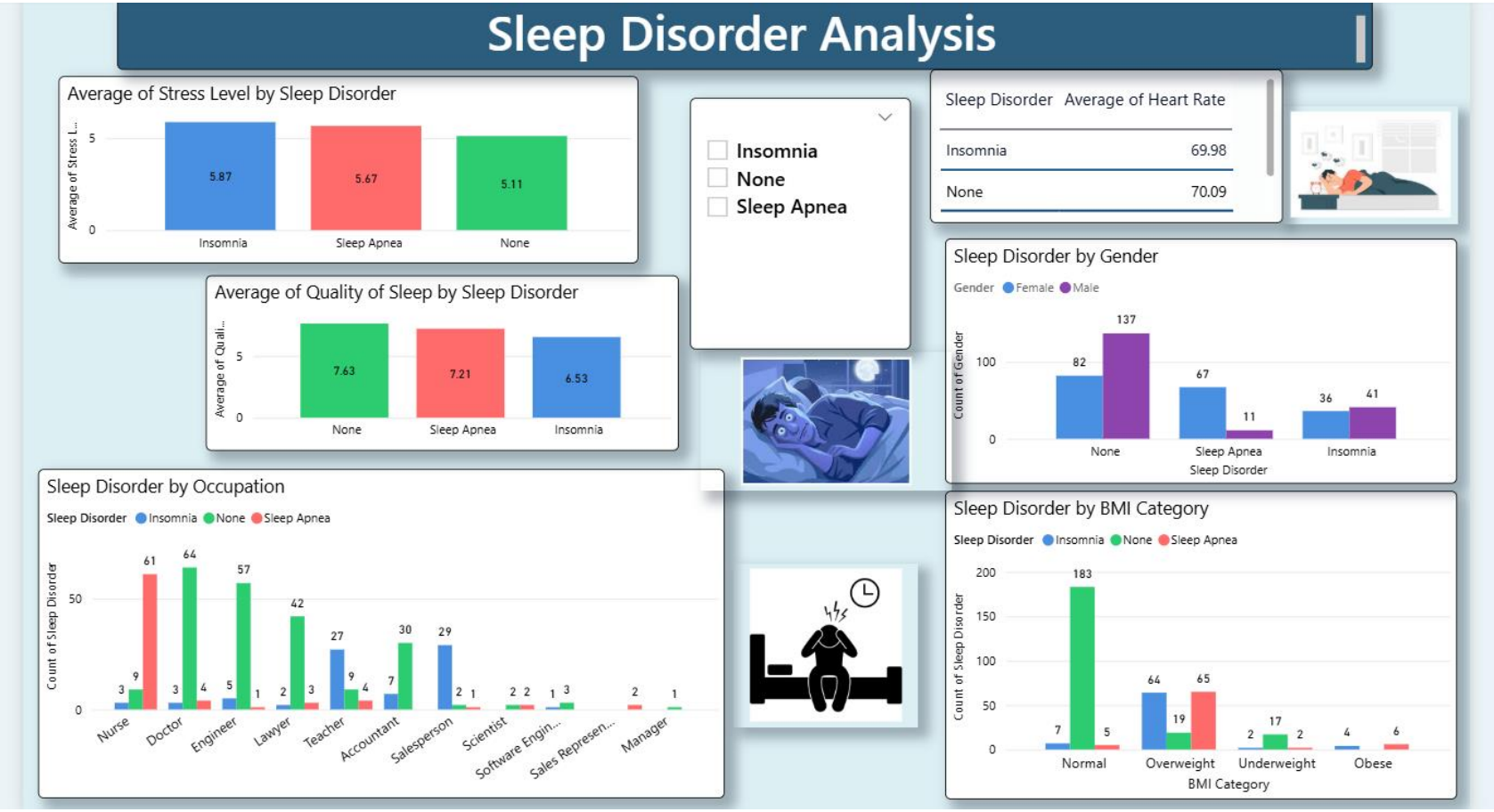
Occupation-based analysis of stress levels, sleep duration, sleep quality, BMI, and sleep disorders. [Link]



# RESULT:

**Sleep Disorder Analysis:**

Analysis of sleep disorders based on stress levels, sleep quality, gender, occupation, and BMI. [Link]



# CONCLUSION:

This project successfully analyzes the relationship between lifestyle factors and sleep health using an interactive Power BI dashboard. The analysis highlights how factors such as stress level, physical activity, BMI, occupation, and gender influence sleep duration, sleep quality, and sleep disorders.

The dashboard effectively provides clear visual insights that help identify patterns and trends related to sleep health. Interactive features like filters and slicers allow users to explore the data dynamically.

**Challenges**

- Managing and organizing multiple variables in the dataset
- Designing clear and readable visualizations
- Selecting appropriate charts to represent relationships

**Potential Improvements**

- Use larger or real-time health datasets
- Integrate predictive analytics for sleep disorder prediction
- Deploy the dashboard online for wider accessibility

This solution demonstrates how data visualization and analytics can support better understanding of sleep health patterns.

# FUTURE SCOPE:

The current project provides valuable insights into sleep health using data visualization. In the future, the system can be enhanced with advanced technologies and additional data sources to provide deeper analysis and more accurate insights.

- Integrate larger and more diverse health datasets to improve analysis accuracy.
- Implement machine learning models to predict sleep disorders and sleep quality.
- Incorporate real-time health data from wearable devices such as smartwatches and fitness trackers.
- Develop a web-based or cloud-based dashboard for easier access and sharing.
- Add additional health factors like diet, screen time, and mental health indicators.
- Enable personalized sleep recommendations based on user lifestyle patterns.
- Improve dashboard interactivity with advanced filters and drill-down analysis.
- Integrate the system with healthcare applications to support medical research and wellness monitoring.

# REFERENCES:

**Sleep Health and Lifestyle Dataset** – Kaggle
https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset

**Microsoft Power BI Documentation** – Microsoft Learn
https://learn.microsoft.com/power-bi

**Power BI Data Visualization Best Practices** – Microsoft Documentation

**National Sleep Foundation** – Research and information on sleep health
https://www.sleepfoundation.org

**World Health Organization (WHO)** – Information on sleep and health research
https://www.who.int

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# Thank You